

The EU Carbon Border Adjustment Mechanism and China's Strategic Response: Carbon Pricing Realignment and Industrial Decarbonization

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Abstract—

I. INTRODUCTION

- A. From Free-Riding to Trade-Linked Enforcement: Why CBAM Matters
- B. CBAM's Particular Relevance for China
- C. Research Question, Scope, and Roadmap

II. LITERATURE REVIEW AND THEORETICAL FRAMEWORK

- A. Climate Clubs and Border Carbon Adjustments
- B. Two-Level Games: Linking International Pressure to Domestic Reform
- C. Carbon Pricing Convergence and Policy Diffusion
- D. Gaps in Existing Research
- E. Analytical Framework

III. METHODOLOGY AND DATA

A. Data Sources

Analysis draws on four primary datasets. Plant-level steel capacity and technology data come from the GEM Iron & Steel Tracker (Global Energy Monitor, 2026), comprising 458 operating plants with unit-level BOF/EAF technology classification, provincial location, and production capacity. Cement capacity data come from the GEM Cement and Concrete Tracker (Global Energy Monitor, 2025), covering 1,016 operating plants with clinker and cement capacity disaggregated by plant type (integrated vs. grinding-only). Sectoral production benchmarks draw on International Energy Agency (2023) and World Steel Association (2024). Carbon pricing data for the CN-ETS are sourced from MEE official publications (Ministry of Ecology and Environment, 2021, 2023); EU ETS price ranges reflect 2024–2025 market data.

B. Emission Factor Derivation

Three sector-specific emission factors are applied throughout:

BF-BOF steel: 2.10 tCO₂/t crude steel, representing Scope 1 direct process emissions per IEA benchmarks (International Energy Agency, 2023), consistent with near-exclusive reliance on coking coal as blast furnace reductant across China's 599 operating blast furnaces.

EAF steel: 0.44 tCO₂/t crude steel, derived from 580 kWh/t average electricity consumption for Chinese EAF operations (above the global benchmark of ≈ 480 kWh/t due to higher DRI and cold pig iron charging) at the MEE national grid emission factor of 0.581 kgCO₂/kWh (Ministry of Ecology and Environment, 2023), yielding 0.337 tCO₂/t Scope 2, plus 0.10 tCO₂/t Scope 1 process emissions. China's power generation mix and the derivation of this grid factor are shown in Fig. 1.

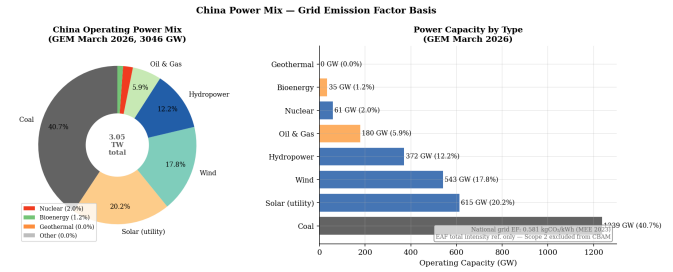


Fig. 1. China's power generation mix and associated grid emission factor (0.581 kgCO₂/kWh) (Ministry of Ecology and Environment, 2023), forming the Scope 2 emissions basis for EAF CBAM liability calculations.

Cement: 0.627 tCO₂/t finished cement, derived from the GEM-observed clinker-to-cement ratio of 0.760 multiplied by the IPCC (2006) clinker emission factor of 0.825 tCO₂/t clinker (calcination: 0.525; coal combustion kiln heat: 0.300) (IPCC, 2006).

C. CBAM Liability Model

Net CBAM liability per tonne of exported product i is computed as:

$$\text{CBAM Liability}_i = \max(0, EF_i \times (P_{\text{EU}} - P_{\text{CN}})) \quad (1)$$

where EF_i is embedded emissions intensity (tCO₂/t), P_{EU} is the prevailing EU ETS allowance price (EUR/tCO₂), and P_{CN} is the verified CN-ETS carbon cost paid in the country of origin. Steel and cement sectors entered CN-ETS coverage in 2024; P_{CN} therefore reflects the prevailing CN-ETS allowance price for covered facilities, currently EUR 10–13/tCO₂. Sensitivity analysis spans two EU ETS price scenarios (EUR 65 and EUR 80/tCO₂) and five CN-ETS price states.

D. Scope and Limitations

Aluminium is excluded from quantitative analysis due to the absence of GEM unit-level smelting data for China. Aggregate CBAM exposure estimates are bounded to current CBAM sectoral scope (HS Chapter 72 iron and steel; cement clinker); downstream embedded emissions in manufactured goods are noted but not modelled. Provincial export volumes are approximated from capacity shares rather than trade flow data, introducing uncertainty in absolute cost estimates.

Causality is a further limitation: the correlation between CBAM implementation and CN-ETS reform acceleration is consistent with the analytical framework in Section II.E, but formal causal identification requires counterfactual evidence beyond the scope of this study. Findings are interpreted as associations informed by theoretical priors, not demonstrated causal effects. These parameters feed directly into the liability model applied in Section V.

IV. CBAM DESIGN AND TRANSMISSION MECHANISMS

A. Policy Design and Transmission Logic

CBAM (Regulation EU 2023/956) entered its definitive phase in January 2026, requiring EU importers to surrender certificates proportional to verified embedded emissions. The mechanism’s strategic significance for China lies in its deduction clause: carbon costs already paid in the country of origin reduce CBAM liability euro for euro, creating a direct fiscal incentive to strengthen domestic carbon pricing and retain that revenue domestically rather than transfer it to the EU budget. Three transmission channels follow from this design: (i) the **direct cost channel**, raising landed cost of Chinese exports proportionally to emissions intensity; (ii) the **revenue retention incentive**, converting potential EU transfers into domestically retained CN-ETS compliance costs; and (iii) the **signalling channel**, in which the EU ETS price acts as an external benchmark for CN-ETS ambition. These channels are tested empirically in Sections V and VI.

V. EMPIRICAL RESULTS

A. Sectoral Exposure: China’s CBAM Vulnerability

China constitutes the single largest non-EU source country for CBAM-liable exports, with exposure concentrated in steel and cement.

Steel. China’s GEM-tracked steel industry comprises 458 plants with 1,005 Mtpa of operating crude steel production capacity—approximately 54% of global crude steel production (World Steel Association, 2024). Of this capacity, 835.7 Mtpa (83.1%) is attributable to BF-BOF operations and 169.5 Mtpa (16.9%) to EAF. China’s EAF share relative to global peers is shown in Fig. 2. China’s weighted-average emissions intensity across its current production mix is $0.831 \times 2.10 + 0.169 \times 0.44 = 1.82 \text{ tCO}_2/\text{t}$, vs. $0.44 \text{ tCO}_2/\text{t}$ for a pure EAF producer—a 4.1:1 ratio that translates directly into the core CBAM competitiveness liability.

Li et al. (2026) estimate aggregate CBAM-induced export losses rising from EUR 0.78 billion under the current pilot sectors to EUR 11.14 billion with full CBAM expansion

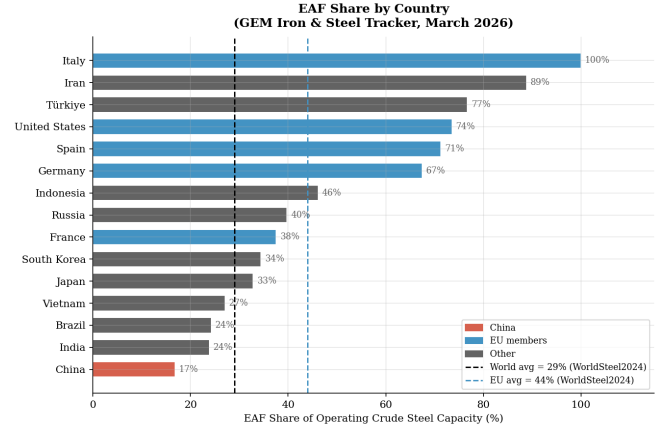


Fig. 2. EAF share of crude steel production capacity (%) by country. China’s 16.9% compares with $\approx 29\%$ globally and $\approx 44\%$ in the EU, and $>70\%$ in the United States (World Steel Association, 2024). Data: GEM Iron & Steel Tracker, March 2026 (Global Energy Monitor, 2026).

across all covered products. Pan et al. (2025) project CBAM costs of EUR 1,157 million by 2034 for China’s iron, steel, and aluminium exports to the EU, with over 65% of the burden falling on Hebei, Jiangsu, Shanxi, Liaoning, and Zhejiang. Provincial BF-BOF capacity distribution underlying this geographic concentration is shown in Fig. 3.

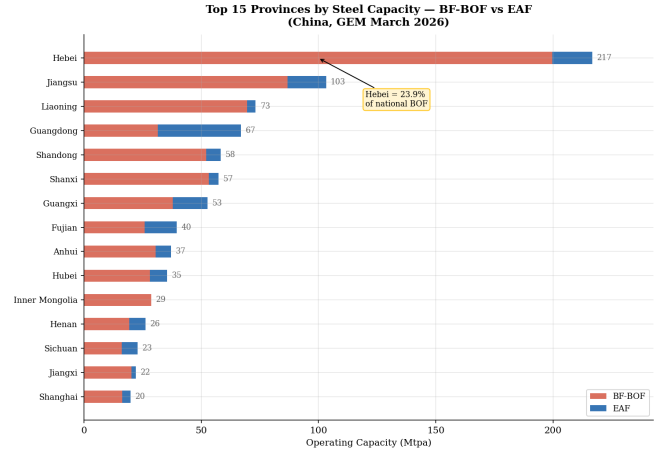


Fig. 3. Operating BOF and EAF capacity (Mtpa) by province, China. Hebei (199.7 Mtpa BOF), Jiangsu (86.8), Liaoning (69.5), Shanxi (53.2), and Shandong (52.0) collectively account for 55.2% of national BOF capacity. Data: GEM Iron & Steel Tracker, March 2026 (Global Energy Monitor, 2026).

Cement. China accounts for approximately 50–51% of global cement output (Global Cement and Concrete Association, 2024). GEM identifies 1,016 operating plants with 1,772 Mtpa cement capacity and 1,346 Mtpa clinker capacity. At the IPCC (2006) clinker emission factor of $0.825 \text{ tCO}_2/\text{t}$ clinker (IPCC, 2006), Chinese integrated plants emit approximately $0.627 \text{ tCO}_2/\text{t}$ of finished cement. Direct CBAM exposure from cement exports to the EU is currently limited but would enlarge substantially under any downstream construction materials expansion.

B. CBAM Cost Quantification

Applying Eq. 1 with emission factors from Section III-B, and with steel and cement carrying $P_{CN} \approx \text{EUR } 10\text{--}13/\text{tCO}_2$ following their 2024 CN-ETS inclusion, net CBAM liability per tonne is given in Table I and visualized in Fig. 4.

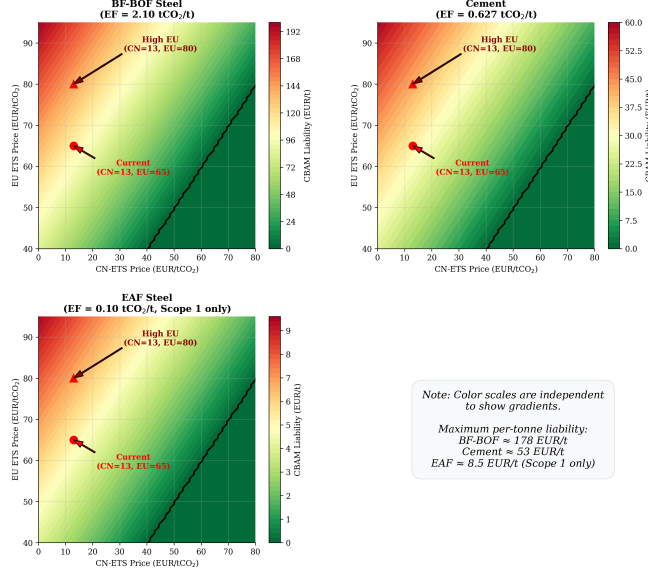


Fig. 4. Heatmap of net CBAM liability (EUR/t) across EU ETS price scenarios and CN-ETS coverage levels for BF-BOF steel, EAF steel, and cement. Derived from Table I.

C. Carbon Price Gap and Convergence Scenarios

As of 2024–2025, the EU ETS price has traded at EUR 50–75/tCO₂ vs. EUR 10–13/tCO₂ for CN-ETS. Three convergence scenarios toward full CBAM offset are illustrated in Fig. 5. Li et al. (2026) find that China’s ETS helps reduce 30–60% of CBAM-induced export losses, with the offset magnitude positively correlated with CN-ETS price levels—underscoring the strategic value of domestic carbon pricing reform as a CBAM mitigation instrument.

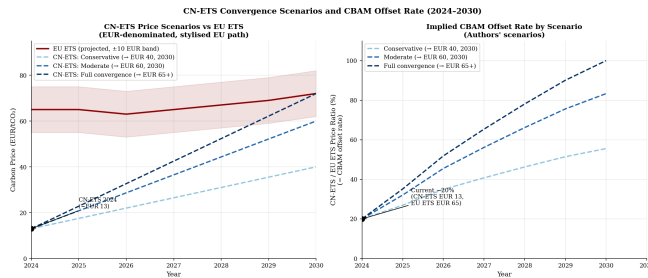


Fig. 5. CN-ETS price convergence scenarios toward EU ETS level, 2024–2035. Conservative: EUR 40/tCO₂ by 2030; Moderate: EUR 60/tCO₂ by 2030; Full convergence: price parity with EU ETS. Authors’ scenarios.

VI. INDUSTRIAL AND POLICY RESPONSE

A. Steel Adjustment Pathways

China’s steel sector presents the most acute CBAM exposure, combining the world’s highest absolute production

volume with an emissions-intensive technology mix geographically concentrated in provinces where decarbonization pathways are structurally constrained (Fig. 3).

The EUR 107.9/t per-tonne liability differential between BF-BOF and EAF (Table I) can be partially neutralized through: (i) product switching, (ii) EAF capacity expansion in coastal provinces, and (iii) process adjustment within BF-BOF via scrap charging or hydrogen injection.

The scrap supply constraint is a binding limit on transition pace. China’s steel fleet is relatively young—reflecting the rapid capacity build-out of the 2000s—and domestic scrap availability remains limited. Chen et al. (2023) project EAF deployment will concentrate in Jiangsu, Guangdong, and Sichuan, while CCUS deployment will concentrate in Hebei, Shandong, Liaoning, and Jiangsu. Scrap availability is projected to rise substantially through 2030–2035 as end-of-life scrap from early 2000s infrastructure reaches recovery (International Energy Agency, 2023), enabling a medium-term decarbonization wave that CBAM accelerates by rewarding early EAF investment.

B. Cement Decarbonization Constraints

Cement decarbonization is constrained by the thermodynamic irreducibility of calcination: $\approx 63\%$ of direct emissions arise from $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$, which cannot be eliminated through fuel switching or energy efficiency alone. GEM identifies 1,016 operating plants with 1,772 Mtpa cement capacity; 782 are fully integrated and 214 are grinding-only. The clinker-to-cement ratio of 0.760 exceeds the global average of 0.73 (Global Cement and Concrete Association, 2024). Translating the cost exposures modelled in Section V-B into investment imperatives, CCS deployment status and CBAM-implied breakeven economics are shown in Fig. 6.

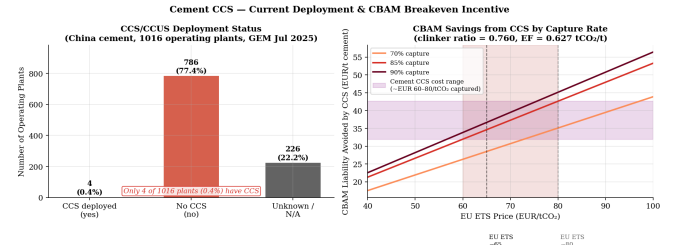


Fig. 6. Cement sector CCS deployment in China and CBAM-implied CCS breakeven cost. Of 1,016 GEM-tracked operating plants, only 4 report active CCS/CCUS (<0.4%). Breakeven assumes 90% CO₂ capture rate (Zhong et al., 2024).

The CBAM cost floor (Section V-B) provides a partial economic signal for CCS investment. With a post-combustion capture rate $\eta = 0.90$ (Zhong et al., 2024), the CBAM saving per tonne of cement from deploying CCS is:

$$\Delta_{\text{CBAM}} = EF_{\text{cement}} \times \eta \times P_{\text{EU}} = 0.627 \times 0.90 \times 65 \approx \text{EUR } 36.7/\text{t cement} \quad (2)$$

CCS becomes economically viable when this saving covers operating costs. At the prevailing CCS cost of EUR 60–80/tCO₂ captured (= EUR 33.7–45.1/t cement),

TABLE I
NET CBAM LIABILITY PER TONNE OF PRODUCT BY CN-ETS COVERAGE AND PRICE SCENARIO (EUR/T)

CN-ETS Scenario	EU ETS = EUR 65/tCO ₂			EU ETS = EUR 80/tCO ₂		
	BF-BOF	EAF	Cement	BF-BOF	EAF	Cement
Not covered (pre-2024 baseline)	136.5	28.6	40.8	168.0	35.2	50.2
Covered, EUR 13/tCO ₂ (<i>current</i>)	109.2	22.9	32.6	140.7	29.5	42.1
Covered, EUR 40/tCO ₂	52.5	11.0	15.7	84.0	17.6	25.1
Covered, EUR 60/tCO ₂	10.5	2.2	3.1	42.0	8.8	12.5
Full parity (EU ETS level)	0.0	0.0	0.0	0.0	0.0	0.0

Note: Emission factors: BF-BOF 2.10 tCO₂/t (International Energy Agency, 2023); EAF 0.44 tCO₂/t (580 kWh/t $\times$ 0.581 kgCO₂/kWh + 0.10; GEM and Ministry of Ecology and Environment 2023); Cement 0.627 tCO₂/t (GEM clinker ratio 0.760 $\times$ IPCC 0.825 tCO₂/t clinker; IPCC 2006). “Pre-2024 baseline” row assumes $P_{CN} = 0$; “current” row reflects CN-ETS entry of steel and cement in 2024 at EUR 13/tCO₂. Figures rounded to one decimal place.

the breakeven EU ETS price is approximately EUR 65–80/tCO₂—placing current EU ETS levels at the lower bound of CCS viability for cement.

The technical floor for clinker content in ordinary Portland cement is 0.65–0.70. China’s primary SCMs—fly ash and blast furnace slag—are byproducts of sectors undergoing structural decline, constraining the clinker substitution pathway and reinforcing the long-run indispensability of CCS.

C. CN-ETS Reform: Before and After CBAM

China launched its national ETS in February 2021, initially covering only the power generation sector ($\approx 40\%$ of total CO₂ emissions) (Ministry of Ecology and Environment, 2021). The intensity-based allocation mechanism does not constrain absolute emissions, and allowance prices have traded in the range of CNY 40–105/tCO₂ (EUR 5–13/tCO₂)—far below EU ETS levels.

MEE has since announced three central reforms aligned with CBAM’s definitive phase: (i) transition from intensity-based to absolute emissions caps by 2027; (ii) sectoral expansion, with steel and cement incorporated into CN-ETS in 2024 and aluminium targeted for inclusion in 2025–2026; and (iii) introduction of allowance auctioning to replace free allocation. Attribution of these reforms solely to CBAM pressure requires caution: the dual carbon goals (*shuang tan*)—peak CO₂ by 2030, neutrality by 2060—provided independent domestic momentum. CBAM is best understood as reinforcing and accelerating an existing trajectory, consistent with the two-level games framework in Section II-B.

D. Political Economy Constraints

Three structural constraints govern the pace of CN-ETS reform and industrial decarbonization.

SOE dominance in steel, cement, and aluminium creates an asymmetric incentive structure: SOEs internalize compliance costs but do not fully capture the export competitiveness benefits of decarbonization, consistently favouring slower reform and more generous free allocation (Nahm & Urpelainen, 2021).

Energy security imperatives constrain the pace of transition away from coal-based steelmaking. Blast furnace operations are integrated with domestic coking coal supply chains

and concentrated in provinces—Hebei, Shanxi, Liaoning—where industrial employment is structurally dependent on BF-BOF production.

Regional heterogeneity in carbon intensity complicates uniform carbon pricing, generating distributional tensions that constrain the political feasibility of stringent absolute caps.

Applied to the two-level game framework (Putnam, 1988), these constraints define China’s domestic win-set. CBAM expands it incrementally by imposing fiscal costs on the status quo, without overriding domestic constraints. Putnam’s framework was developed for democratic states with formalized legislative ratification; China’s win-set is instead shaped through party channels and SOE advisory bodies rather than legislative veto. The framework remains analytically useful as a metaphor for the range of politically feasible outcomes, provided this limitation is acknowledged. The resulting reform trajectory is likely to be gradual, sectoral, and subject to implementation delays relative to MEE’s announced targets.

VII. DISCUSSION

A. Enterprise-Level Strategies for Reducing CBAM Exposure

B. Carbon Market Reform: Closing the Price Gap to Retain Revenue

C. EU–China Cooperation: From Coercive Pressure to Market Alignment

VIII. CONCLUSION

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